

LinearRegression_StudentsPerformanceFactors

March 22, 2026

1 Student Performance Prediction

2 Can study habits and additional factors explain exam scores?

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.ensemble import RandomForestRegressor
```

```
[2]: df_raw= pd.read_csv("StudentPerformanceFactors.csv").copy()
df=df_raw
```

```
[3]: print(df.head())
print(df.info())
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	\
0	23	84	Low	High	
1	19	64	Low	Medium	
2	24	98	Medium	Medium	
3	29	89	Low	Medium	
4	19	92	Medium	Medium	

	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	\
0	No	7	73	Low	
1	No	8	59	Low	
2	Yes	7	91	Medium	
3	Yes	8	98	Medium	
4	Yes	6	65	Medium	

	Internet_Access	Tutoring_Sessions	Family_Income	Teacher_Quality	\
0	Yes	0	Low	Medium	
1	Yes	2	Medium	Medium	
2	Yes	2	Medium	Medium	

3	Yes	1	Medium	Medium
4	Yes	3	Medium	High

	School_Type	Peer_Influence	Physical_Activity	Learning_Disabilities	\
0	Public	Positive	3	No	
1	Public	Negative	4	No	
2	Public	Neutral	4	No	
3	Public	Negative	4	No	
4	Public	Neutral	4	No	

	Parental_Education_Level	Distance_from_Home	Gender	Exam_Score
0	High School	Near	Male	67
1	College	Moderate	Female	61
2	Postgraduate	Near	Male	74
3	High School	Moderate	Male	71
4	College	Near	Female	70

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 6607 entries, 0 to 6606

Data columns (total 20 columns):

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	Hours_Studied	6607 non-null	int64
1	Attendance	6607 non-null	int64
2	Parental_Involvement	6607 non-null	object
3	Access_to_Resources	6607 non-null	object
4	Extracurricular_Activities	6607 non-null	object
5	Sleep_Hours	6607 non-null	int64
6	Previous_Scores	6607 non-null	int64
7	Motivation_Level	6607 non-null	object
8	Internet_Access	6607 non-null	object
9	Tutoring_Sessions	6607 non-null	int64
10	Family_Income	6607 non-null	object
11	Teacher_Quality	6529 non-null	object
12	School_Type	6607 non-null	object
13	Peer_Influence	6607 non-null	object
14	Physical_Activity	6607 non-null	int64
15	Learning_Disabilities	6607 non-null	object
16	Parental_Education_Level	6517 non-null	object
17	Distance_from_Home	6540 non-null	object
18	Gender	6607 non-null	object
19	Exam_Score	6607 non-null	int64

dtypes: int64(7), object(13)

memory usage: 1.0+ MB

None

3 Initial data inspection

The dataset contains 6,607 observations and 20 columns. It includes both numerical and categorical variables related to student habits, school environment, and previous academic performance.

The target variable is `Exam_Score`.

```
[4]: df.isna().sum()
```

```
[4]: Hours_Studied          0
Attendance                 0
Parental_Involvement       0
Access_to_Resources        0
Extracurricular_Activities  0
Sleep_Hours                0
Previous_Scores            0
Motivation_Level           0
Internet_Access            0
Tutoring_Sessions          0
Family_Income              0
Teacher_Quality            78
School_Type                0
Peer_Influence             0
Physical_Activity          0
Learning_Disabilities       0
Parental_Education_Level   90
Distance_from_Home         67
Gender                     0
Exam_Score                 0
dtype: int64
```

4 Checking missing values

```
[41]: df['Teacher_Quality'].fillna(df['Teacher_Quality'].mode()[0], inplace=True)
df['Parental_Education_Level'].fillna(df['Parental_Education_Level'].mode()[0],
    ↪inplace=True)
df['Distance_from_Home'].fillna(df['Distance_from_Home'].mode()[0],
    ↪inplace=True)
```

```
[6]: df.isna().sum()
```

```
[6]: Hours_Studied          0
Attendance                 0
Parental_Involvement       0
Access_to_Resources        0
Extracurricular_Activities  0
Sleep_Hours                0
```

```

Previous_Scores      0
Motivation_Level     0
Internet_Access      0
Tutoring_Sessions    0
Family_Income        0
Teacher_Quality      0
School_Type          0
Peer_Influence       0
Physical_Activity     0
Learning_Disabilities 0
Parental_Education_Level 0
Distance_from_Home   0
Gender               0
Exam_Score           0
dtype: int64

```

4.1 Handling missing values

Some columns contained a small number of missing values (less than 2% of the dataset).

- Categorical variables (Teacher_Quality, Parental_Education_Level, Distance_from_Home) were filled using the most frequent value (mode).

Since the percentage of missing data was low, no rows were removed.

5 Encoding categorical variables

```
[7]: df_encoded = pd.get_dummies(df, drop_first=True)
```

```
[43]: df_encoded.head()
```

```

[43]:   Hours_Studied  Attendance  Sleep_Hours  Previous_Scores  Tutoring_Sessions  \
0             23           84             7             73             0
1             19           64             8             59             2
2             24           98             7             91             2
3             29           89             8             98             1
4             19           92             6             65             3

      Physical_Activity  Exam_Score  Parental_Involvement_Low  \
0                   3           67                   True
1                   4           61                   True
2                   4           74                   False
3                   4           71                   True
4                   4           70                   False

      Parental_Involvement_Medium  Access_to_Resources_Low  ...  \
0                   False                   False  ...
1                   False                   False  ...

```

2	True	False	...
3	False	False	...
4	True	False	...

	Teacher_Quality_Medium	School_Type_Public	Peer_Influence_Neutral	\
0	True	True	False	
1	True	True	False	
2	True	True	True	
3	True	True	False	
4	False	True	True	

	Peer_Influence_Positive	Learning_Disabilities_Yes	\
0	True	False	
1	False	False	
2	False	False	
3	False	False	
4	False	False	

	Parental_Education_Level_High_School	\
0	True	
1	False	
2	False	
3	True	
4	False	

	Parental_Education_Level_Postgraduate	Distance_from_Home_Moderate	\
0	False	False	
1	False	True	
2	True	False	
3	False	True	
4	False	False	

	Distance_from_Home_Near	Gender_Male
0	True	True
1	False	False
2	True	True
3	False	True
4	True	False

[5 rows x 28 columns]

6 Simple statistics.

```
[9]: print(f'Shape of data {df.shape}')
      print(df.describe().round(2))
```

Shape of data (6607, 20)

	Hours_Studied	Attendance	Sleep_Hours	Previous_Scores \
count	6607.00	6607.00	6607.00	6607.00
mean	19.98	79.98	7.03	75.07
std	5.99	11.55	1.47	14.40
min	1.00	60.00	4.00	50.00
25%	16.00	70.00	6.00	63.00
50%	20.00	80.00	7.00	75.00
75%	24.00	90.00	8.00	88.00
max	44.00	100.00	10.00	100.00

	Tutoring_Sessions	Physical_Activity	Exam_Score
count	6607.00	6607.00	6607.00
mean	1.49	2.97	67.24
std	1.23	1.03	3.89
min	0.00	0.00	55.00
25%	1.00	2.00	65.00
50%	1.00	3.00	67.00
75%	2.00	4.00	69.00
max	8.00	6.00	101.00

6.1 Descriptive statistics

The average exam score is around 67 points. Students studied about 20 hours on average, while average attendance was close to 80%. The numerical variables appear to be on different scales, but linear regression can still be used as a baseline model.

7 Choosing data for examining.

8 Splitting data for training and testing.

```
[10]: X=df_encoded.drop("Exam_Score", axis=1)
      y=df_encoded["Exam_Score"]

      X_train, X_test, y_train, y_test = train_test_split(X,y, test_size=0.2,
      ↪random_state=42)
```

8.1 Train-test split

The data was split into training and testing sets using an 80/20 ratio. The training set was used to fit the model, while the test set was used to evaluate predictive performance on unseen data.

```
[30]: lr_model =LinearRegression()
      lr_model.fit(X_train, y_train)

      y_pred=lr_model.predict(X_test)
      r2 = r2_score(y_test, y_pred)
      print(f"{r2:.2f}")
```

```
coefficients = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': lr_model.coef_
}).sort_values(by='Coefficient', ascending=False)

print("Predicted exam results", y_pred.round(2))
```

0.77

Predicted exam results [64.53 65.27 71.53 ... 66.17 63.42 71.76]

```
[ ]: coefficients.head(10).round(2)
```

```
[ ]:
           Feature  Coefficient
20  Peer_Influence_Positive      1.05
13  Internet_Access_Yes         0.96
25  Distance_from_Home_Near      0.85
10  Extracurricular_Activities_Yes  0.58
19  Peer_Influence_Neutral       0.57
23  Parental_Education_Level_Postgraduate  0.51
4   Tutoring_Sessions           0.51
24  Distance_from_Home_Moderate     0.32
0   Hours_Studied               0.29
1   Attendance                  0.20
```

Some features, such as distance from home, appear to have a relationship with exam scores.

However, this does not necessarily imply causation. These variables may be indirectly related to other underlying factors.

8.2 Model results

The linear regression model achieved an R^2 score of approximately 0.77.

This indicates that the model explains about 77% of the variance in exam scores.

The result suggests that the selected features, including encoded categorical variables, provide a good representation of factors influencing student performance.

```
[33]: print(f"Coefficient: {lr_model.coef_.round(2)}")
      print(pd.DataFrame (lr_model.coef_, X.columns, columns=["Coefficient"]).round(2))
      print(f"Intercept Coefficient: {lr_model.intercept_.round(2)}")
      print(f"Result R^2 on the test set: {lr_model.score(X_test, y_test): .2f}")

      print("Sample exam results:", lr_model.predict(X_test).round())
```

```
Coefficient: [ 0.29  0.2  -0.01  0.05  0.51  0.19 -2.   -1.07 -2.1  -1.03  0.58
-1.04
-0.55  0.96 -1.11 -0.64 -1.02 -0.56  0.02  0.57  1.05 -0.86 -0.46  0.51
0.32  0.85 -0.03]
```

	Coefficient
Hours_Studied	0.29
Attendance	0.20
Sleep_Hours	-0.01
Previous_Scores	0.05
Tutoring_Sessions	0.51
Physical_Activity	0.19
Parental_Involvement_Low	-2.00
Parental_Involvement_Medium	-1.07
Access_to_Resources_Low	-2.10
Access_to_Resources_Medium	-1.03
Extracurricular_Activities_Yes	0.58
Motivation_Level_Low	-1.04
Motivation_Level_Medium	-0.55
Internet_Access_Yes	0.96
Family_Income_Low	-1.11
Family_Income_Medium	-0.64
Teacher_Quality_Low	-1.02
Teacher_Quality_Medium	-0.56
School_Type_Public	0.02
Peer_Influence_Neutral	0.57
Peer_Influence_Positive	1.05
Learning_Disabilities_Yes	-0.86
Parental_Education_Level_High_School	-0.46
Parental_Education_Level_Postgraduate	0.51
Distance_from_Home_Moderate	0.32
Distance_from_Home_Near	0.85
Gender_Male	-0.03
Intercept	Coefficient: 41.92
Result	R^2 on the test set: 0.77
Sample exam results:	[65. 65. 72. ... 66. 63. 72.]

8.3 Model interpretation

- If study time increases by 1 hour, the model predicts an increase of approximately 0.29 points.
- If attendance increases by 1%, the predicted score increases by about 0.20 points.
- One additional tutoring session is associated with an increase of about 0.51 points.
- More physical activity is associated with a small increase (~0.15 points).

Each effect is relatively small, but combined they can significantly increase the predicted exam score.

Example: - 5 hours of studying → +1.45 points - 3 tutoring sessions → +1.53 points - 10% higher attendance → +2.0 points

Total predicted increase: +4.98 points

```
[ ]: plt.scatter(y_test, y_pred, alpha=0.6)
min_val = min(y_test.min(), y_pred.min())
```

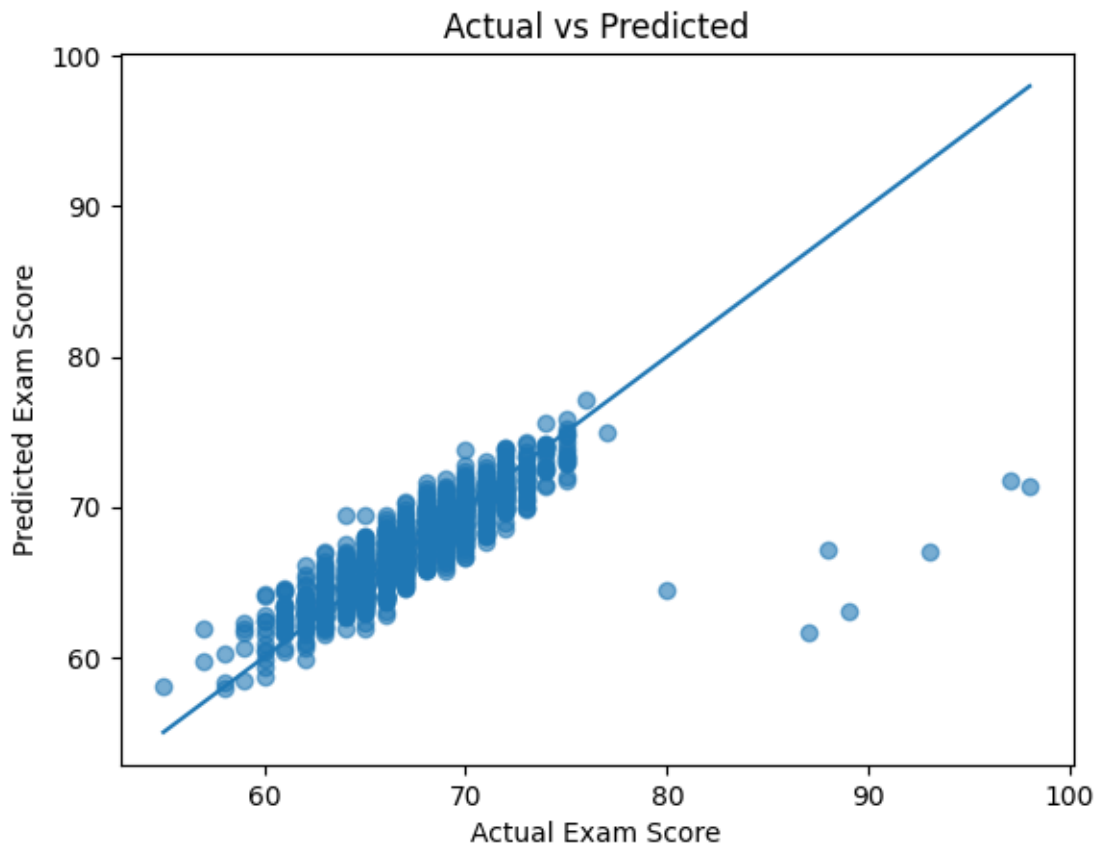


```

max_val = max(y_test.max(), y_pred.max())

plt.plot([min_val, max_val], [min_val, max_val])
plt.xlabel("Actual Exam Score")
plt.ylabel("Predicted Exam Score")
plt.title("Actual vs Predicted")
plt.show()

```



The plot shows actual vs predicted exam scores. Most points are close to the line, which means the model predicts quite well. The R^2 score is about 0.77, so the model fits the data well. The model slightly underestimates higher exam scores.

8.4 Additional statistics

```

[34]: from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
      from sklearn.metrics import (
          mean_absolute_error,
          mean_squared_error,
          root_mean_squared_error,
          r2_score
      )

```

```
)
```

```
[38]: r2_test = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = root_mean_squared_error(y_test, y_pred)

print('\n=== Model quality statistics ===')
print(f'R2 (test)      : {r2_test: .2f}')
print(f'MAE (Mean Abs. Error) : {mae: .2f}')
print(f'MSE (Mean Sq. Error)   : {mse: .2f}')
print(f'RMSE (Mean Sq. Error) : {rmse: .2f}')
```

```
=== Model quality statistics ===
R2 (test)      : 0.77
MAE (Mean Abs. Error) : 0.45
MSE (Mean Sq. Error)   : 3.26
RMSE (Mean Sq. Error) : 1.80
```

8.5 Model performance

The model achieved an R^2 score of 0.77, indicating a good fit. The average prediction error (MAE) is about 0.45 points, which means predictions are very close to actual exam scores. The RMSE value of 1.80 also confirms that prediction errors are relatively small.

```
[ ]: # Residual analysis
# (The difference between the actual exam results and the model's assessment)
residuals = y_test - y_pred
print('\n=== Residual analisis (forecast errors)===')
print(f'Mean residual      : {residuals.mean():.2f}')
print(f'Std of residuals   : {residuals.std():.2f}')
print(f'Minimum residual    : {residuals.min():.2f}')
print(f'Maksimum residual   : {residuals.max():.2f}')
```

```
=== Residual analisis (forecast errors)===
Mean residual      : -0.05
Std of residuals   : 1.80
Minimum residual    : -1.57
Maksimum residual   : 28.01
```

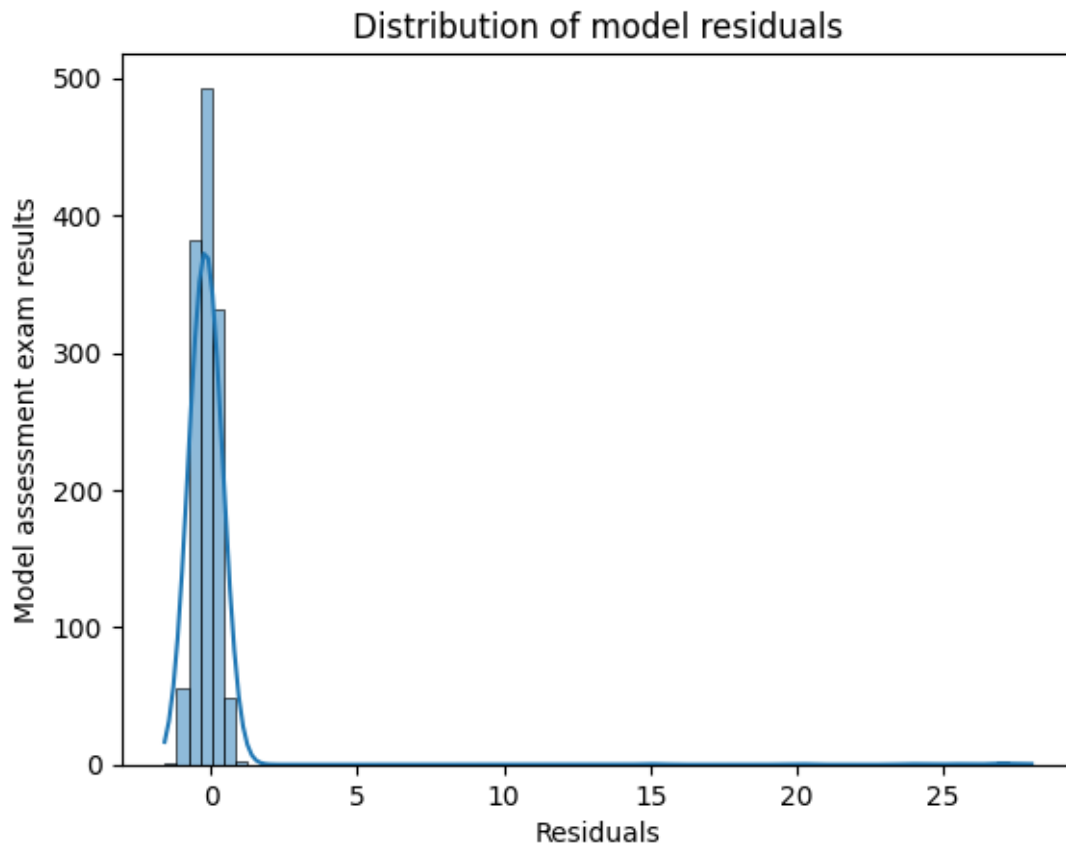
Residual analysis

The average error is close to zero, which means the model is not biased. Most prediction errors are small (around 1–2 points).

However, there are some larger errors, which suggests that the model struggles with certain cases.

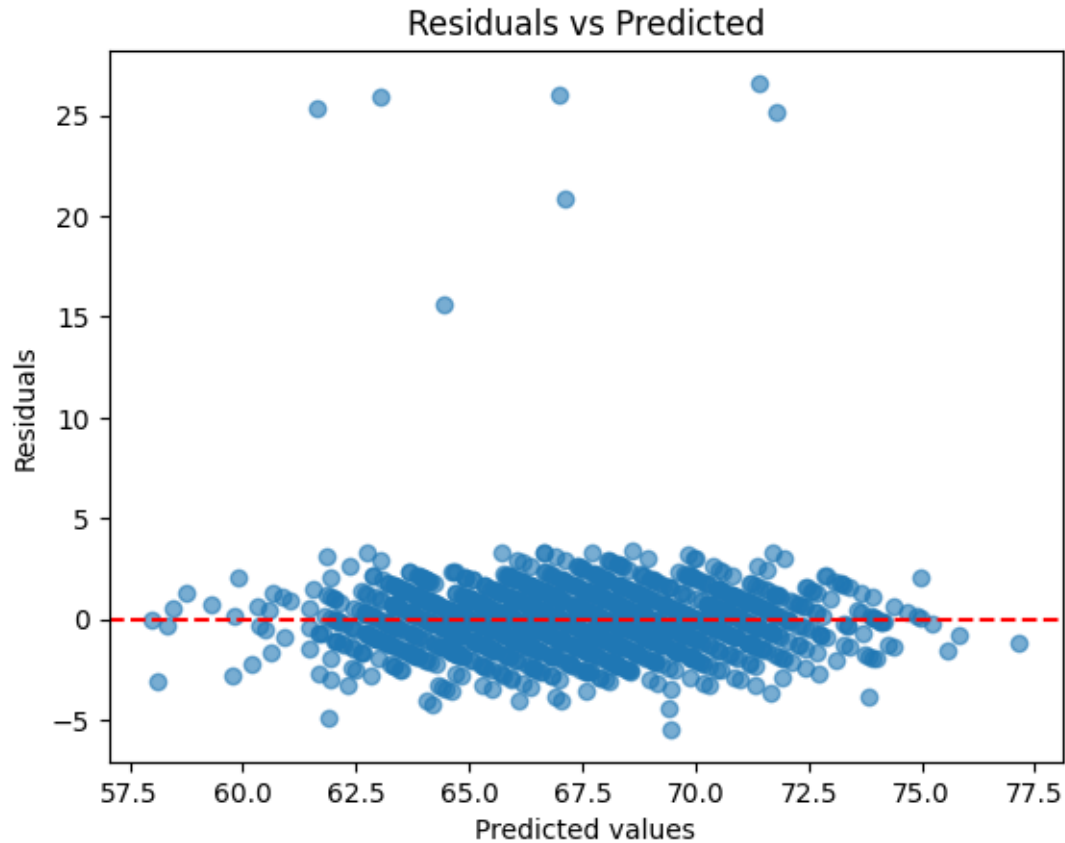
Residuals histogram plot

```
[40]: sns.histplot(residuals,kde=True)
plt.title("Distribution of model residuals")
plt.xlabel("Residuals")
plt.ylabel("Model assessment exam results")
plt.show()
```



The residuals are approximately normally distributed and centered around zero, which suggests that the linear regression model fits the data reasonably well. A few large positive residuals indicate some outliers where the model underestimates exam scores.

```
[ ]: plt.scatter(y_pred, residuals, alpha=0.6)
plt.axhline(0, color="red", linestyle="--")
plt.xlabel("Predicted values")
plt.ylabel("Residuals")
plt.title("Residuals vs Predicted")
plt.show()
```



Residuals are mostly centered around zero with no strong pattern, indicating a reasonably good model fit. A few large positive residuals suggest that the model underestimates some observations.

8.6 Final conclusion

The results show that exam scores are influenced not only by study time, but also by other factors. In addition to studying, elements such as attendance, tutoring, and student environment are linked to better results. This means that while studying is important, other factors can also significantly impact the final exam score. The model achieved a good result ($R^2 = 0.77$), which confirms that these factors together explain student performance well.